

California Native Plant–Pollinator Dataset

Biodiversity & Native Planting Planner — Data Handoff README

Honest framing: This dataset covers confirmed and likely **bee and butterfly** pollinator interactions with native/observed plant species across **55 of California's 58 counties**. **Hummingbird coverage was investigated and found to be absent** from the underlying data sources used here — this is identified below as a known gap and a target for future enhancement, not something this dataset currently supports.

1. Project Context

Project: Biodiversity & Native Planting Planner

Problem statement: Urban gardeners lack simple, automated visual tools to identify and choose native plant species that support local pollinator populations.

Role of this dataset: This file is a data-analyst deliverable, prepared to be consumed by the IT automation team as the backing data for a visual tool. It is not itself the visual tool — no dashboard, map, or UI is included here.

2. What This Dataset Is

This dataset links California plant occurrence records to pollinator interaction records, so that a given plant species can be associated with the bee or butterfly species known or likely to interact with it, along with the county in which that plant was observed.

Metric	Value
Total rows	3,084
Total columns	14
Unique plant species	79
Unique counties represented	55 of 58 California counties
Pollinator groups covered	Bee, Butterfly
Pollinator groups NOT covered	Hummingbird (see Section 5 — Known Limitations)
Bee interaction rows	2,618
Butterfly interaction rows	466
Observation years represented	2025–2026

3. Data Sources & Pipeline

Source 1 — GBIF (Global Biodiversity Information Facility): plant occurrence records filtered to Kingdom = Plantae, Country = United States, State = California, with coordinates present. This provided the base set of observed plant species and where they were seen.

Data source: [Link](#)

Citation: GBIF.org (22 July 2026) GBIF Occurrence Download <https://doi.org/10.15468/dl.65fcts>

Source 2 — GloBI (Global Biotic Interactions API): queried once per unique plant scientific name from the cleaned GBIF list, to retrieve known ecological interactions involving that plant.

Processing steps applied, in order:

- Cleaned and deduplicated the GBIF plant list; removed records missing coordinates or scientific names; retained species-level taxonomic rank.
- Filtered GBIF's basisOfRecord field to **HUMAN_OBSERVATION records only**, in Google Colab, before extracting unique plant names for GloBI querying. Other record types (e.g. PRESERVED_SPECIMEN) were excluded at this stage and are not represented anywhere in the final dataset.
- Queried the GloBI API once per unique plant species (8,412 species) to retrieve interaction records.
- Filtered interactions to those taxonomically identified as bees (families Apidae, Halictidae, Megachilidae, Andrenidae, Colletidae), butterflies (superfamily Papilionoidea and family Hesperidae), or hummingbirds (family Trochilidae), using each interaction's full taxonomic path rather than common name text alone.
- Combined two evidence tiers: interactions explicitly typed as pollination behavior (pollinatedBy, flowersVisitedBy, visitedBy) and a broader recovered set from GloBI's generic interactsWith type, narrowed to the same taxonomic filter, to increase species and county coverage.
- Corrected a taxonomic string-matching error that had misclassified some plant-to-plant interactions as butterfly interactions (the plant subfamily “Papilionoideae” was a substring match for the butterfly superfamily “Papilionoidea”); affected rows were identified and removed.
- Derived county and state from each record's latitude/longitude via offline reverse geocoding, since GBIF's own free-text locality field was populated for only a small fraction of records.
- Removed records that geocoded outside California (e.g., border points resolving to Baja California, Mexico).
- Deduplicated to one row per unique plant species × pollinator species × county combination.

4. Coverage Detail

Geographic coverage is uneven and reflects real GBIF observation density, not sampling choices made in this pipeline. Coastal and Bay Area counties (Santa Clara, San Mateo, Monterey, Marin, Santa Cruz, San Diego) have the most records; several rural counties (Amador, Glenn, Colusa, Modoc, Kings) have only 1–3 rows each. A county appearing in the dataset does not guarantee broad species choice within it — some counties are represented by a single plant–pollinator pair.

Species coverage (79 species) reflects the subset of California's plant occurrence records that also had a documented or inferred bee/butterfly interaction in GloBI. The great majority of GBIF's ~8,400 unique California plant species had no matching interaction record and are not represented here.

evidence_type distinguishes two confidence tiers within this dataset: **confirmed_pollination** (306 rows) comes from GloBI interaction types that explicitly describe pollination or flower-visiting behavior.

general_interaction (2,778 rows) comes from GloBI's broader interactsWith type, narrowed to the same bee/butterfly taxonomic filter, but describing ecological co-occurrence rather than confirmed pollination behavior.

5. Known Limitations

- **No hummingbird coverage.** Hummingbird (Trochilidae) interactions were explicitly searched for at every stage of this pipeline, including the broader interactsWith recovery pass. Zero rows survived in either pass. This reflects a genuine sparsity of hummingbird interaction data in GloBI for California plants — it is not a filtering error. If hummingbird support is required for launch, it will need a supplementary source (for example, curated native hummingbird-plant lists from Calscape or the Xerces Society), merged in and clearly labeled as a separate source.
- **Uneven county coverage.** 55 of 58 counties are represented, but row counts per county range from over 200 to as few as 1. Sparse counties may return very few or no results in the tool.
- **EstablishmentMeans and Study_title have been removed from the final file** (they were 100% empty across all 3,084 rows in every earlier version, so keeping them added no information). establishmentMeans was GBIF's native/introduced/cultivated status field; study_title was GloBI's source-study citation field. Neither was populated at the source for any record in this dataset.
- **Native status is inferred, not verified.** Because establishmentMeans was empty, this dataset does not contain a confirmed native/introduced flag. Every plant here was simply observed within California — that is not the same as being confirmed native to California, and a small number of introduced or cultivated species could be present without a way to identify them from this data alone. Verifying true native status against an authoritative source (e.g., Calscape's California native plant database) is listed as a future enhancement in Section 6.
- **BasisOfRecord is not carried into the final file, but was used as a cleaning filter.** GBIF occurrence records were filtered to HUMAN_OBSERVATION only, in Google Colab, before species names were extracted for GloBI querying. Every row in this dataset reflects a human observation; preserved specimens and other record types were excluded upstream and are not represented. The column itself was dropped from the final output since all rows already meet this single criterion — see the Data Dictionary's "Columns Removed From This File" section for details.
- **General_interaction rows are a broader signal than confirmed pollination.** These increase coverage but represent ecological co-occurrence (e.g., logged together in a study) rather than directly observed pollination behavior in every case.
- **Species-level identification only.** Records identified only to genus or family level in GBIF were excluded, which improves match reliability but further reduces the species count.

6. Recommended Next Steps (Future Enhancement)

- Source a supplementary hummingbird-plant interaction dataset (e.g., Calscape, Xerces Society) and merge it in as a clearly labeled additional evidence source.
- Cross-reference the 79 plant species in this dataset against an authoritative native-species list (e.g., Calscape) to add a verified native/introduced flag, since establishmentMeans could not provide one.
- Investigate whether additional GBIF filters (e.g., relaxing basisOfRecord or taxonRank constraints) can recover more species without compromising data quality.
- Consider weighting or filtering tool recommendations by evidence_type so that confirmed_pollination rows can be presented with higher confidence than general_interaction rows, if the tool's design calls for that distinction.

- Revisit sparse counties (fewer than ~10 rows) to decide whether the tool should show a “limited data” message rather than an empty result for those areas.

7. Files in This Handoff

File	Description
california_pollinator_plants.csv	The final dataset described in this document.
DATA_DICTIONARY.pdf	Column-by-column definitions, types, and notes.
Data_Generation_Process_README.pdf	This document.
California_pollinator_plants_v2.ipynb	Data cleaning and API pipeline with python using google colab

Prepared by: Data Analyst — Biodiversity & Native Planting Planner project.